HW2 多元線性回歸 — AI 協助對話紀錄

工具：Claude Code（VS Code 擴充功能）｜專案：物聯網 HW2 多元線性回歸（CRISP-DM）｜對話時間：2026-10-27 21:16 ～ 2026-10-28 09:56｜9 則使用者訊息、38 則 AI 回覆
本檔由完整對話紀錄擷取：保留使用者與 AI 的文字內容，省略工具呼叫、程式輸出、截圖與系統提示。時間為 UTC+8。

使用者 2026-10-27 21:16

幫我閱讀 @describtion.md 並看一下目前專案做到哪裡了。
另外我想要我的網站在github.io 上面部署，不是streamlit

Claude 2026-10-27 21:18

Project status is clear. Next I'll check one detail: how the current Streamlit app handles data, so the static version behaves the same.

Claude 2026-10-27 21:19

Now I'll check the math against statsmodels/sklearn on the real data before building the UI, using macOS's built-in JavaScript engine since Node isn't installed:

Claude 2026-10-27 21:21

Now the main app logic. It covers the seven tabs, the dataset dropdown, custom URL/upload, and the live simulator:

Claude 2026-10-27 21:24

Cleaning up a few rough spots before testing:

Claude 2026-10-27 21:24

Now an end-to-end test in headless Chrome, served over a local HTTP server the same way GitHub Pages serves it:

Claude 2026-10-27 21:24

The first tab loads with no JS errors. To exercise every tab and all three datasets, I'll use a temporary test harness that drives the page in an iframe (deleted afterwards):

Claude 2026-10-27 21:25

All 12 dataset × method combinations now run cleanly across all 7 tabs. P2 drops to 17 features, which still meets the >13 requirement. Next, a visual check with screenshots:

Claude 2026-10-27 21:26

The collinearity note now names the right columns. I'll tidy the wording, remove the test harness, and update the README for GitHub Pages:

Claude 2026-10-27 21:26

Now the README: I'll replace the Streamlit sections with GitHub Pages instructions.

Claude 2026-10-27 21:26

The site is now a static web page that GitHub Pages can host, and it no longer uses Streamlit. I tested it in headless Chrome: all three datasets work with all four feature-selection methods, across all 7 tabs. It isn't on GitHub yet, because the folder isn't a git repository so far.

What's already done (from before)

Requirement	Status
Kaggle dataset with >13 features	✅ WHO Life Expectancy, 20 features
Feature selection and model evaluation	✅ RFE / SelectKBest / Lasso; R^2, RMSE, MAE
Prediction plot with intervals	✅ 95% confidence and prediction intervals
CRISP-DM write-up	✅ in README.md
+1 bonus: dropdown with two problems	✅ adds a second problem, King County house prices
+2 bonus: any dataset URL	✅
Exported GPT chat PDF	❌ not in the project
NotebookLM summary	⚠️ the text exists, but nothing shows it came from NotebookLM

What I changed

- **Rewrote the site in HTML + JavaScript.** Pages can't run Python, so the regression math now runs in the browser: OLS, the 95% intervals, VIF, the three feature-selection methods and the assumption tests.
  - The results match statsmodels/sklearn to more than 10 significant digits.
  - Files: index.html, assets/stats.js, assets/app.js, assets/style.css.
- **Kept the bonus features.** The dropdown still switches between the two problems. The custom-dataset option takes a URL as before, and now also accepts an uploaded CSV

file.

- **Fixed a bug in the second problem.** In the King County data, `sqft_living` equals `sqft_above` + `sqft_basement`, so the regression can't be solved. The site now drops `sqft_basement` automatically and says so on the Data Preparation tab. That leaves 17 features, still above the 13 required.
- **Corrected the method:**
  - Missing values are now filled using only the training data, so the test set doesn't leak into training.
  - The Durbin–Watson independence test now runs in the data's original order. On the WHO data it now flags autocorrelation, because each country appears in several years.
- **Added:**
  - A table comparing the four feature-selection strategies.
  - A chart showing how Lasso chose its penalty.
  - The share of test points that fall inside the prediction interval: 94.2% for a 95% interval.
  - A single-feature regression chart with interval bands.
- **Updated** README.md with local preview steps and deployment steps, and added `.nojekyll`.

What you need to do

1. **Publish to GitHub Pages.** Create a repository on GitHub and push this folder to it; the commands are in the README. Then go to Settings → Pages and choose `main` / `(root)`. The site will be at `https://<your-account>.github.io/<repo>/`. I can run `git init` and make the first commit if you want.
2. **Replace the NotebookLM summary with your real one.** Run NotebookLM yourself and paste its summary into `ai()` in assets/app.js and into the README. The assignment requires it to come from NotebookLM.
3. **Add the GPT chat PDF.** Export it and put it in `docs/`; the AI tab says it's there.
4. **Optional:** app.py (the old Streamlit version) and `streamlit` in requirements.txt are no longer needed. You can delete them, or keep them as the Python version. analysis.py still works as an offline script.

To preview locally, run `python3 -m http.server 8000` and open http://localhost:8000. Opening `index.html` directly won't load the CSV files.

使用者 2026-10-28 09:23

Apply this visual style to the user's existing project. Inspect and preserve its stack, copy, routes, and behavior. Follow existing component and styling conventions. Treat framework, library, and page examples in these design

instructions as illustrations; they do not require a stack change or new sections. Scope changes to the user's requested work.

Keep controls keyboard accessible, focus visible, and text legible. Preserve reading order, responsive behavior, and reduced-motion preferences. Verify actual states at the rendered width. The user's stack and content take precedence over illustrative implementation details.

Sketch

Identity

Use a shared working sheet with warm paper, handwritten text, pencil-like drawings, blue notes, red corrections, and yellow highlights. Slightly uneven lines and taped references give the page a drawn quality, while consistent spacing keeps it easy to read.

Layout

Open with a large handwritten statement beside a smaller cluster of notes and a simple line drawing. An irregular underline and a hand-drawn arrow can direct attention toward the main action. Facts, when useful, sit beneath a drawn bracket.

Arrange longer reading as open thoughts with generous space between them. Features can use small line illustrations beside notes, with a little vertical staggering on wide layouts. Editorial images look like taped references, and a process can sit on a slightly darker irregular paper strip with circled numbers and connecting arrows. Quotes can use outlined speech notes; comparisons can use broad horizontal rows with one highlighted sheet. End with a large red statement above a lightly divided footer.

Use only the content groups the brief calls for. The working-sheet treatment does not require team doodles, fabricated comments, prices, or a fixed sequence of sections.

Typography

Use Working Sheet Marker for large statements and headings, bound to "Sketch Working Marker". Use Working Sheet Hand for reading, bound to "Sketch Working Hand". These are separate original drawn fonts. Keep the marker face heavier and the hand face large enough for continuous reading; its base is intentionally larger than a typical sans body face. If these font files are unavailable, choose a legible handwritten pair with the same weight distinction and reassess the line breaks.

Keep lines loosely spaced and headings slightly irregular. Avoid tight typesetting that makes the handwriting run together. Small rotations belong to selected notes or headings, not every paragraph.

Color and surfaces

Warm paper carries dark pencil ink. Muted ink supports secondary text; erased-line grey supplies quiet rules. Blue marks annotations, red marks actions and emphasis, and pale yellow supplies highlights. Lighter scraps distinguish a few notes and image mounts.

Use faint paper grain, uneven thin strokes, irregular corner curves, and small hard offsets. Tape can have torn ends and partial transparency. Keep the page mostly open paper, with each scrap or highlight serving a clear reading purpose.

Components

Buttons have irregular outlined edges and a small pencil or red offset shadow. Hover can straighten or tilt them slightly while compressing the shadow; press settles onto the page. Use a clear dashed blue focus outline.

Features and benefits pair small drawings or checks with text. Optional comparison rows use yellow behind the featured choice, while selection changes the action's treatment separately. Disclosures use hand-drawn marks with an obvious open state. Keep real button, link, and disclosure semantics beneath the drawn appearance.

Images

Use natural-color photographs or artwork inside pale paper mounts, with small rotations and a tape strip near the edge. Contain artwork when its margins matter; crop scenes around the subject. Preserve the photograph's detail instead of converting every image into a drawing. Reserve dimensions, provide useful alternative text, and hide decorative tape, arrows, and doodles from assistive technology.

Motion

Use very brief changes for button tilt and pressed depth. Keep handwriting, arrows, paper, and sketches still. Reduced motion removes transforms and transitions while preserving visible state feedback. Do not animate strokes drawing themselves across the page.

Responsive behavior

Start with a single-column working sheet, two-column facts, and two-column footer groups. At 680px, features and references can form two-column groups, facts expand across a row, and comparisons gain a second column. At 1050px, use the full split opening, staggered reference row, horizontal process, and wider comparison ledger. On small screens, reduce rotations and decorative offsets, let notes stack, and keep handwritten reading text generously sized.

Design limits

Avoid illegible cursive, heavy crumpled-paper textures, dense doodle borders, and polished geometric cards. Drawn marks should direct attention or group content. They must not obscure text, alter the reading order, or make controls look unfinished.

Authoritative tokens

CSS variable	Value
--dp-color-ground	#f7f2e7
--dp-color-ink	#38352f
--dp-color-ink-muted	#514d44
--dp-color-line	#b3ac9d
--dp-color-accent	#9e3a2e
--dp-color-annotation	#344c69
--dp-color-highlight	#f0dfa0
--dp-color-scrap	#fcf8ef
--dp-typography-reading-face	"Sketch Working Hand", cursive
--dp-typography-display-face	"Sketch Working Marker", cursive
--dp-typography-body	24px
--dp-typography-reading	23px
--dp-typography-detail	21px
--dp-spacing-item-gap	16px
--dp-radius-control	9px 4px 17px 5px / 4px 12px 5px 9px
--dp-border-fine-stroke	1.5px

CSS variable	Value
--dp-motion-control	100ms

Claude 2026-10-28 09:24

I'll restyle the existing static site in place. The structure, copy, controls and behavior stay the same; only the CSS, the font link, a few decorative marks, and the chart colors change.

The "Sketch Working Hand/Marker" font files don't exist here, and the copy is Traditional Chinese, so I'm using a legible handwritten pair from Google Fonts. LXGW WenKai TC handles the Chinese and Kalam the Latin, with heavier weights for headings. First the stylesheet:

Claude 2026-10-28 09:25

Now the font link and the hand-drawn underline in the header:

Claude 2026-10-28 09:25

Next, the charts and markup in app.js: pencil/blue/red chart palette, handwritten chart fonts, the CRISP-DM process strip, and a yellow highlight on the current method's comparison row.

Claude 2026-10-28 09:25

Two more edits: support a row class in the table helper, and wait for the handwritten fonts before drawing the first charts so Plotly measures the right glyphs.

Claude 2026-10-28 09:25

Now I'll verify the rendered result at desktop and phone widths:

Claude 2026-10-28 09:26

Desktop looks right. The one defect is that chart legends now collide with the x-axis titles, because the handwritten font is larger. I'll check mobile before fixing it:

Claude 2026-10-28 09:26

Mobile stacks correctly and stays readable. Now the legend fix: legends anchor to the bottom of the whole figure instead of the plot area. I'm also making interval values a bit smaller on phones:

Claude 2026-10-28 09:26

The legends are clear now. Last, I'll check the keyboard focus ring:

Claude 2026-10-28 09:26

The dashed blue focus ring shows. Cleaning up the test harness and server:

Claude 2026-10-28 09:26

The site now has the hand-drawn "working sheet" style. I checked it in headless Chrome at 1400px and at 390px phone width. The text, controls, tabs and regression results are the same as before.

Fonts: The design's "Sketch Working Hand/Marker" fonts don't exist here, and they wouldn't cover Chinese anyway. I used a readable handwritten pair from Google Fonts instead: Kalam for Latin letters and numbers, and LXGW WenKai TC (霞鶩文楷) for Chinese. Headings and numbers use the bold weight, reading text the regular one. The original font names stay first in the font list, so they'll be used if you ever add those files.

What changed:

- **Header:** the big handwritten title has a red hand-drawn underline. The subtitle sits beside it as a slightly tilted note with a strip of yellow tape.
- **Colours and paper:** warm paper with a faint grain, pencil-ink text, blue for notes, red for actions and emphasis, yellow for highlights. All values come from the design's colour list.
- **Panels and cards:** pale paper scraps with uneven corners and a small hard shadow. The number boxes are tilted very slightly, and stop tilting on phones.
- **Buttons:** uneven outline with a red offset shadow. Hover tilts the button a little, and pressing pushes it down onto the page.
- **Keyboard focus:** a dashed blue outline on every control.
- **CRISP-DM steps:** a darker paper strip with circled numbers and arrows. It runs across on wide screens and stacks on phones.
- **Tables and labels:** the active tab gets a red uneven underline, and section headings have a yellow highlighter stroke. In the strategy comparison table, the row for the current method is highlighted yellow.
- **Charts:** recoloured to match, with handwritten chart text. The first charts are drawn after the fonts load, so the text lines up.
- **Layout:** one column below 680px, two-column chart groups from 680px, and the side panel sits beside the content from 1050px.
- **Reduced motion:** button tilting and transitions are off. The hover shadow change still shows.

Fixed while checking: with the larger font, chart legends overlapped the x-axis titles. Legends now sit at the bottom of each chart. On phones, the long interval values in the number boxes are a bit smaller.

The changes are in assets/style.css, index.html (font link and the drawn underline) and assets/app.js (chart colours and fonts, the step strip and the highlighted row).

使用者 2026-10-28 09:31

網站風格沒有動誒，你只有在大標題放了個波浪

Claude 2026-10-28 09:31

A big wave under the title with the rest of the page unchanged points to a stale cache: your browser got the new `index.html` (which added the wave SVG) but is still using the old cached `style.css`. Without the new CSS, the wave has no size limit, so it renders large. I'll check whether you've already pushed to GitHub:

Claude 2026-10-28 09:31

The style files on disk are already the new version; your browser was most likely still using the old cached `style.css`. That's why the big wave appeared: `index.html` had loaded the new version with the wave, but without the new CSS the wave has no size limit, so it shows up large. The screenshots I took in headless Chrome showed the full restyle (warm paper, handwritten fonts, red/blue marks, scrap-paper cards), so the files themselves are fine.

I added version numbers to the CSS/JS links in index.html (`style.css?v=2` and so on), so the browser has to fetch the new files.

Please:

1. Reload with a hard refresh: **Cmd + Shift + R** on Mac.
2. If it still looks the same, make sure you're opening this folder (`HW2/index.html`) through a local server (`python3 -m http.server 8000`, then http://localhost:8000), not another copy.

If it's still wrong after that, send me a screenshot and I'll look at it.

When you change CSS/JS later, bump the number (`?v=3`). GitHub Pages also caches files, so the same thing can happen after you deploy.

使用者 2026-10-28 09:32

https://github.com/BlueET1/AIot-HW2-MLR-web-.git
幫我push 到這裡

Claude 2026-10-28 09:32

I'll check whether the GitHub repository already has content before pushing:

Claude 2026-10-28 09:33

The push was rejected with HTTP 400, which often happens on a first push larger than git's default HTTP buffer (about 2.4 MB here). I'll raise the buffer for this repo and retry:

Claude 2026-10-28 09:33

The push worked. I'll replace the placeholder site link in the README with the real address, and check whether I can turn on GitHub Pages from here:

Claude 2026-10-28 09:33

Pages is turned on. I'll wait for the first build to finish and confirm the site loads:

Claude 2026-10-28 09:33

The build is done. I'll load the live page in headless Chrome to confirm the analysis actually runs:

Claude 2026-10-28 09:34

The project is on GitHub and the site is live on GitHub Pages. I opened the live page in headless Chrome: it loads the data and finishes the analysis ("已完成建模：2,928 筆樣本、20 個特徵，RFE 選出 14 個").

- **Repository:** https://github.com/BlueET1/AIot-HW2-MLR-web-
- **Website:** https://blueet1.github.io/AIot-HW2-MLR-web-/

What I did:

- Pushed the whole project to the `main` branch. `.venv` is excluded by `.gitignore`.
- The first push failed with HTTP 400 because the push was larger than git's default upload buffer. I raised the buffer for this repo only, and the retry went through.
- Turned on GitHub Pages using the `gh` CLI, which is logged in as BlueET1, with source `main` / `(root)`.
- Replaced the placeholder site link at the top of README.md with the real address, in a second commit.

This machine had no git user name or email configured, so I set them for this repo only. Commits show as `BlueET1 <a23144408@gmail.com>`; tell me if you'd prefer a different name.

Still to do:

1. Replace the NotebookLM summary with one you actually got from NotebookLM. It's in `ai()` in assets/app.js and in the README.

2. Put the exported GPT chat PDF in the `docs/` folder; the AI tab says it's there.
3. After editing CSS/JS, bump the `?v=2` in index.html to `?v=3`. Otherwise browsers may keep the old cached file, as happened with the style earlier.

To update the site later, run `git add . && git commit -m "..." && git push`. Pages redeploys automatically in about a minute.

使用者 2026-10-28 09:42

我要怎麼給 NotebookLM 網路解法研究摘要？給我中文prompt

Claude 2026-10-28 09:42

NotebookLM only answers from the sources you add to a notebook, so first add some web write-ups that solve the same problem, then give it the prompt.

Step 1: Add sources (5–10 is enough)

Create a new notebook and use **「探索來源」(Discover sources)**. Type:

```
WHO Life Expectancy dataset Kaggle multiple linear regression feature selection
```

It will find related articles for you to add. You can also paste links via **新增來源 → 網站** (Add source → Website). Good search terms:

- `kumarrajarshi life expectancy who regression notebook`
- `life expectancy WHO linear regression multicollinearity VIF`
- `life expectancy prediction Lasso RFE feature selection`

Kaggle notebook pages sometimes don't import completely. Medium articles, GitHub READMEs, and blog posts work more reliably.

Step 2: Prompt (paste it into the NotebookLM chat box)

```
請根據我提供的所有來源，針對「WHO Life Expectancy（全球預期壽命）資料集的多元線性回歸分析」撰寫一份繁體中文研究摘要，用於大學作業報告。

要求：
1. 長度 200～300 字，以一段完整的段落呈現，不要條列。
2. 內容須涵蓋以下面向（若來源中有提到）：
   - 各解法中對預期壽命影響最顯著的特徵（正相關與負相關各舉例），以及其係數方向
   - 資料前處理的常見做法（缺失值處理、類別變數編碼、極端值或對數轉換）
   - 共線性問題（哪些特徵之間高度相關、是否使用 VIF 檢查）
   - 使用的特徵選擇方法（如 Lasso、RFE、逐步迴歸、相關係數篩選）及其效果
   - 模型評估結果（R²、RMSE 等常見數值範圍）以及是否檢驗回歸假設
3. 最後用一句話總結這些網路解法的共通結論或不足之處。
4. 只根據來源內容撰寫，不要自行補充來源中沒有的數據；若各來源說法不一致，請指出差異。
5. 在摘要後另外列出你主要引用的來源標題。
```

Step 3: Use the result

1. Copy NotebookLM's answer into the 【NotebookLM 摘要】 section of `ai()` in assets/app.js, and into the NotebookLM section of README.md.
2. Take a screenshot of the NotebookLM page, or export it as a PDF. You can put it in `docs/` as proof that it came from NotebookLM.

Once you have the summary, paste it here and I'll put it in both places and push it to GitHub.

使用者 2026-10-28 09:47

@notebooklm.md 幫我更新到網站和readme上面

Claude 2026-10-28 09:47

The text is in both places. I'll commit and push:

Claude 2026-10-28 09:47

The push was rejected because the remote has commits I don't have, so nothing was deployed (the `0` is the old file). I'll look at what's there before merging:

Claude 2026-10-28 09:47

The remote commit is a README edit made on GitHub, which touches the same area as mine. I'll rebase my commit on top of it so their edit is kept:

Claude 2026-10-28 09:48

NotebookLM 摘要已更新到網站和 README，並已推上 GitHub。我在線上確認過，網站裡已經是新的摘要文字。

- **網站**：AI 協助 / NotebookLM 分頁裡的 【NotebookLM 摘要】 已換成你的內容（assets/app.js）。
- **README**：同一段摘要也放進去了（README.md）。
- **文字處理**：內容照 notebooklm.md 原文，只做兩個調整：
  - 把引用編號被刪掉後留在中文字之間的多餘空白去掉，例如「插值法 之差異」改成「插值法之差異」。
  - 把 `**…**` 和 `\(R^2\)` 這些標記改成純文字 R^2。
- **快取版號**：index.html 的版號改成 `?v=3`，瀏覽器才會抓到新檔。

第一次推送被拒絕，因為你在 GitHub 上改過 README（那次叫 "Refine README content and structure"）。我沒有強制覆蓋，而是把我的更新接在你的修改後面，所以你的修改都保留了。

注意：你在 GitHub 上改過的 README，把「加分項目」的說明刪掉了（原本寫著下拉選單兩個 Problem 加 1 分、自訂 CSV 加 2 分）。評分標準有這兩項加分，如果你不是故意要拿掉，建議補回去；要的話跟我說，我幫你加。

另外，README 裡還有 AI 對話 PDF 要放進 `docs/` 這一項還沒做。

使用者 2026-10-28 09:49

我要怎麼輸出我跟你的對話

Claude 2026-10-28 09:49

有三種方法，由簡單到彈性：

1. 用 `/export` 指令（最簡單）
在對話輸入框打 `/export`，可以把這段對話存成文字檔，或複製到剪貼簿。我不確定你這個 VS Code 版本的細節，指令沒反應的話用方法 2。拿到文字檔後，用 Chrome 開啟、列印、選「另存為 PDF」，或貼到 Word / Google Docs 再匯出 PDF。

2. 讓我幫你產生（推薦）
這個專案的完整對話紀錄在你電腦上：

`/Users/chenzhiheng/.claude/projects/-Users-chenzhiheng-Desktop----------HW2/` 資料夾裡的 `.jsonl` 檔。

跟我說一聲，我可以把它轉成排版好的 Markdown 或 HTML，再用 Chrome 輸出成 PDF，放進專案的 `docs/` 資料夾並推上 GitHub。我會只留下你和我的文字對話，把工具呼叫、截圖等雜訊拿掉。

3. 直接用 VS Code 的對話面板
把對話內容整段選取複製，貼到文件裡再匯出 PDF。長對話這樣比較不整齊。

有兩點要先確認：

- **作業要求寫的是匯出 ChatGPT 的對話。** 你這次主要是用 Claude Code 做的，建議問一下老師或助教，Claude 的對話紀錄可不可以代替，或是要同時附上。我們網站和 README 的「GPT 輔助內容」文字也寫的是 GPT，如果改交 Claude 紀錄，我可以幫你把那段改成「AI 輔助（Claude Code）」。
- **對話裡有你的 Email、電腦資料夾路徑和 GitHub 帳號。** 交出去之前看一下，要的話我可以先把這些遮掉。

要我用方法 2 幫你做嗎？

使用者 2026-10-28 09:53

好

Claude 2026-10-28 09:53

我先看你剛打開的那個匯出檔，也看一下對話紀錄在哪裡：

Claude 2026-10-28 09:53

你已經自己匯出了 `.txt` 和 `.pdf`，所以不用再轉換。我先檢查裡面有沒有個人資料，再決定怎麼放：

Claude 2026-10-28 09:55

對話紀錄已經放進專案並推上 GitHub，網站上的 PDF 連結我開過，可以正常打開。

你已經自己匯出了 `.txt` 和 `.pdf`（24 頁），所以我沒有另外轉檔，只做了這幾件事：

- **整理檔案**：原本檔名有空格和拼字錯誤（histroy），我改名後放進 `docs/`：
  - docs/HW2-chat-history.pdf
  - docs/HW2-chat-history.txt
- **加連結**：網站「AI 協助 / NotebookLM」分頁和 README.md 都加了這兩個檔案的連結。
- **檢查內容**：裡面沒有密碼或金鑰，只有你的 Email 和本機資料夾路徑。Email 本來就在 GitHub 的 commit 紀錄裡，所以我沒有遮掉。如果你想遮掉，跟我說，我處理後重新推上去。

這份匯出是你在 09:52 做的，所以只包含到那個時間點為止的對話，不包含之後我們的對話。要交最新版，可以再用 `/export` 匯出一次，覆蓋 `docs/` 裡的檔案。

兩件事還沒處理：

1. **作業要求的是 ChatGPT 對話。** 網站和 README 的標題仍寫「GPT 輔助內容」，但連結的是 Claude 的紀錄。如果老師接受 Claude 的對話，我建議把標題改成「AI 輔助（Claude Code）」；要改的話跟我說。如果要求一定是 ChatGPT，就需要另外附 ChatGPT 的對話。
2. **README 的「加分項目」說明還是被刪掉的狀態。** 評分有 +1（下拉選單兩個 Problem）和 +2（自訂資料集）兩項加分，要不要補回去？

使用者 2026-10-28 09:56

改成剛剛的方法二，你幫我做

Claude 2026-10-28 09:56

我來從完整紀錄重新產生一份：只留你和我的文字對話，拿掉工具呼叫和截圖，再輸出 PDF。先看紀錄格式：